Computer Networks Lab

Cisco Packet Tracer — Assignment 3

Logical Network Segmentation

All practicals must be completed inside **Cisco Packet Tracer**.
This assignment requires exactly **two screenshots** as submission.

Tool: Cisco Packet Tracer

How to Submit

- Complete the practical entirely inside **Cisco Packet Tracer**. No other tool is accepted.
- Save your file as `RollNo_Set3.pkt` (e.g. `20221234_Set3.pkt`) and submit it.
- Along with the file, submit exactly **two screenshots** as described at the end of the assignment.
- Screenshots must be clear and show all device labels and relevant details.

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Scenario

Elevate Corp's **Sales** and **Human Resources (HR)** teams currently share a single switch, placing both departments in the same broadcast domain — a growing security and traffic concern for IT Manager Mr. Arjun Kapoor. He wants the two departments **logically separated**, yet still able to communicate with each other when necessary (e.g. for payroll data), rather than being fully isolated. The challenge is that the solution must work using the existing switch along with the **only one free router interface** currently available. Design a network setup that achieves this controlled logical separation.

Department Plan:

Group ID	Group Name	Network	Gateway
10	Sales	192.168.10.0/24	192.168.10.1
20	HR	192.168.20.0/24	192.168.20.1

Device Addressing Plan:

Device	Group	IP Address	Default Gateway
Sales-PC1	10	192.168.10.10	192.168.10.1
Sales-PC2	10	192.168.10.20	192.168.10.1
HR-PC1	20	192.168.20.10	192.168.20.1
HR-PC2	20	192.168.20.20	192.168.20.1
Subnet Mask: `255.255.255.0` on all PCs			

Switch Port Allocation:

Switch Port	Connected To	Group
Fa0/1, Fa0/2	Sales-PC1, Sales-PC2	10
Fa0/3, Fa0/4	HR-PC1, HR-PC2	20
Gi0/1	Router (single available interface)	Both

Submission

Submit the following **two screenshots**:

1. The completed **topology** in Packet Tracer Logical View showing the router, switch, and all four PCs connected, with the switch's group/port configuration visible (e.g. via the relevant CLI output or configuration panel) and IP labels visible on all PCs.
2. A **ping result** in the Command Prompt of Sales-PC1 successfully reaching HR-PC1 across the two groups, showing zero packet loss.

Note: Think carefully about how a single physical link and a single router interface can still serve **two different gateway addresses** at the same time. The switch-side link and the router-side configuration must agree on how traffic from each group is identified as it crosses that one shared connection.